

1 DMC Implementation Guide - Manufacturing Instruction Editor

Purpose

This document describes how to operate the Manufacturing Instruction Editor (MIE).

Requirements

License: DMC-MDL

MPDV-DMC is required.

Designing a production process in the DME-MIE

The main function of the DME-MIE is the graphic modeling of a production process for an existing production model (FactoryModel).

For more information: refer to [DMC_ImplementationGuide.pdf](#) (section 4)

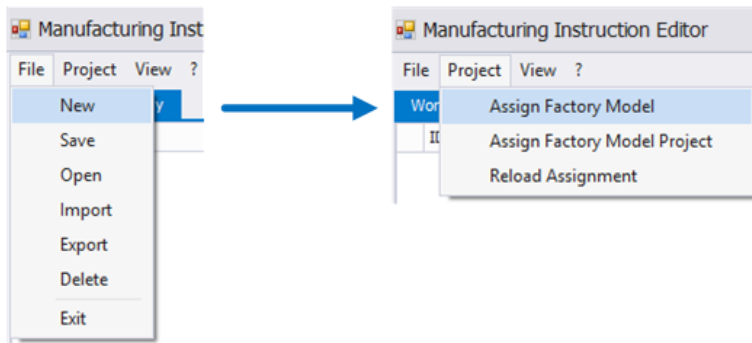
If you want to model a manufacturing instruction, a completed production model should already exist as a DMC factory model.

First step creating a manufacturing instruction: Create a new MIE project to model a manufacturing instruction using MIE. Assign a factory model. In general, you can also model a manufacturing instruction without assigning a factory model. But you should absolutely assign a factory model to avoid misconfigurations and runtime errors. If you assign a factory model, all components and their properties are available for data modeling in the MIE.

To create a new MIE project, click "File" → "New" and assign a unique project name. Then assign an existing factory model created with the FME to the new project. As an alternative, you can also assign a stand-alone factory model (that is not yet part of an FME project). Click "Project" → "Assign factory model project" if you want to assign a factory model that is part of a DME-FME project. Click "Project" → "Assign factory model" if you want to assign a stand-alone factory model (one that is not yet part of an FME project).



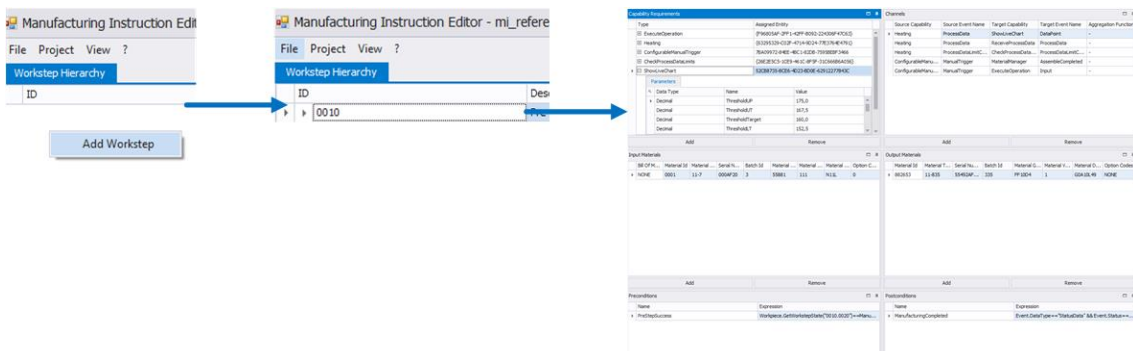
Please note: If you make changes to the assigned factory model, you have to reload this factory model in the MIE project. To do so, click "Project" → "Reload assignment".



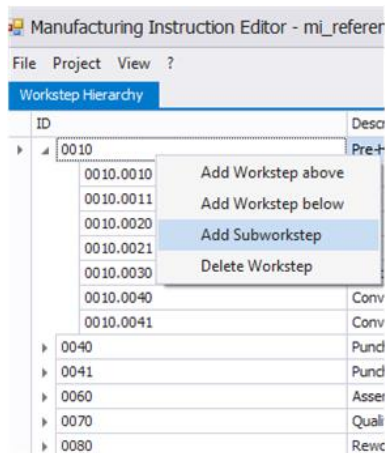
You can start modelling production processes, once you have created a project and assigned a factory model to the project. Assigning the first process step:

- Go to the "Workstep hierarchy"
 - Open the context menu by right-clicking
 - Select the option "Add work step" Assign a unique ID including description in the configuration dialog.
- Then go to "Assignment" and assign the process step to a component of the factory model. The "Workstep hierarchy" shows the new process step. Now you have to configure the process step.

Use the following views to configure the process step: "Capability requirements", "Channels", "Input materials", "Output materials", "Preconditions" and "Postconditions". At first select the created process step. Edit the parameters according to your processes in the above-mentioned views. Use the option "Add" to add new parameters or the option "remove" to delete parameters. To edit existing parameters, double-click the required parameter and change its data in the dialog.



Select the required process step in the "Workstep hierarchy" if you want to add a work step to this process step. Then right-click this process step to open the context menu and select "Add subworkstep" (work step). Create and configure the work step like a process step. If a process step includes several work steps, the "workstep hierarchy" displays these work steps in groups. You can use the same context menu to add new process steps. You can add a new process step above or below an existing process step. To do so, click "Add workstep above" or "Add workstep below". Open the context menu and use the option "Delete workstep" to delete process steps or work steps.



To complete data modeling, just add new process steps and work steps to your model as described above. Configure the new process and work steps according to your process.